

# Energy Transfer as an Observable Coordinate: A Finite Navier–Stokes Bridge from Shell Balances to Certified Readout Inversion

Yaoharee Lahtee

Open Civil Science Initiative, Bangkok, Thailand

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## Abstract

A companion Fourier–Galerkin Navier–Stokes observability programme shows that total- and shell-energy Lie jets can generically determine a finite state modulo spatial translation in several exactly certified finite cases, but it leaves quantitative inversion and noisy recovery open. This note isolates the role of nonlinear energy transfer in those open problems. The main correction is negative but useful: for prescribed forcing, shell transfer  $T = \dot{I} + 2\nu SI - F$  is an invertible affine reparameterization of the first shell-energy jet, so transfer variables do not create observability rank beyond  $(I, \dot{I})$ . Their advantage is instead structural and numerical. They make explicit the internal redistribution that total energy cancels, and they admit a derivative-free finite-window balance with direct interval-error propagation. We also prove a constructive kinematic lemma: every Fourier mode added on the boundary of a cubic cutoff  $N \geq 2$  belongs to a non-collinear Navier–Stokes triad containing a mode from the previous cutoff. This removes lack of triad connectivity as an obstruction to the all-resolution saturation conjecture, while leaving algebraic independence of observable minors open. Finally, the programme’s analytic three-mode subcase is upgraded to an explicit quantitative inverse radius. These are partial closures of the inner-certificate frontier, not a proof of all-resolution observability, global injectivity, continuum regularity, or the Clay Millennium problem.

## 1 Scope and claim discipline

Let  $K_N = \{k \in \mathbb{Z}^3 : 0 < \|k\|_\infty \leq N\}$  be a finite cubic Fourier cutoff for periodic incompressible Navier–Stokes. This note concerns the finite Fourier–Galerkin system and its energy readouts. Standard Navier–Stokes energy identities are used as standard mathematics and are not claimed as original results.

The work is motivated by a Readout Genesis interpretation in which restoration and transmitted difference are distinct roles. That interpretation is not used as a premise for physical energy. In particular, no informational quantity is identified with SI Joules without a separate domain translation. All mathematical claims below are established directly in the finite NS model.

## 2 Shell balance: transfer versus dissipation

For a squared-radius shell label  $s$ , let

$$I_s(t) = \frac{1}{2} \sum_{|k|^2=s} |\widehat{u}_k(t)|^2$$

under the declared Fourier normalization. The finite shell balance can be written

$$\boxed{\dot{I}_s = T_s - 2\nu s I_s + F_s,} \quad (1)$$

where  $T_s$  is the quadratic nonlinear transfer and  $F_s$  is a prescribed shell-energy input. In the unforced closed finite truncation the nonlinearity conserves total kinetic energy, hence

$$\boxed{\sum_s T_s = 0.} \quad (2)$$

Thus the nonlinear term redistributes energy among retained shells while the viscous term removes it. Summing (1) over  $s$  erases the internal transfer pattern by (2); retaining the individual shell equations keeps that pattern visible.

### 3 Transfer is not an extra-rank oracle

Write  $I = (I_1, \dots, I_m)^T$  and  $S = \text{diag}(s_1, \dots, s_m)$ . For known viscosity and prescribed forcing,

$$T = \dot{I} + 2\nu SI - F. \quad (3)$$

**Proposition 1** (First-jet/transfer equivalence). *For prescribed  $F$ , the observation pairs  $(I, T)$  and  $(I, \dot{I})$  have the same local Jacobian rank at every state where the derivatives are defined.*

*Proof.* Equation (3) gives

$$\begin{bmatrix} I \\ T \end{bmatrix} = \begin{bmatrix} I_m & 0 \\ 2\nu S & I_m \end{bmatrix} \begin{bmatrix} I \\ \dot{I} \end{bmatrix} - \begin{bmatrix} 0 \\ F \end{bmatrix}.$$

The block triangular matrix has determinant one and is therefore invertible. Multiplication by an invertible output-coordinate map preserves Jacobian rank.  $\square$

For the exact  $N = 1$  modular witness already used by the observability programme, the companion checker verifies

$$\text{rank } D(I, \dot{I}) = \text{rank } D(I, T) = 5,$$

and verifies the differential transfer dependence  $\sum_s DT_s = 0$ . Hence transfer makes the mechanism visible but does not bypass the shell-jet rank ceiling.

### 4 Derivative-free finite-window transfer

Integrating (1) over  $[t_0, t_1]$  yields

$$\boxed{\int_{t_0}^{t_1} T_s dt = I_s(t_1) - I_s(t_0) + 2\nu s \int_{t_0}^{t_1} I_s dt - \int_{t_0}^{t_1} F_s dt.} \quad (4)$$

This avoids numerical differentiation of measured energy data. If certified absolute interval radii for the four right-hand terms are respectively  $r_1, r_0, r_I, r_F$ , interval arithmetic gives

$$\boxed{\text{rad}\left(\int_{t_0}^{t_1} T_s dt\right) \leq r_1 + r_0 + 2\nu s r_I + r_F.} \quad (5)$$

This closes a specific noise-propagation subproblem: transfer-window uncertainty can be certified without constructing high-order time derivatives. It does not yet provide a certified inverse from those windows to the complete Fourier state.

## 5 All-cutoff kinematic triad connectivity

**Lemma 2** (Boundary-to-old-cutoff triad connection). *For every integer  $N \geq 2$  and every  $k \in K_N$  with  $\|k\|_\infty = N$ , there exist  $p \in K_{N-1}$  and  $q \in K_N$  such that*

$$p + q = k, \quad p \times q \neq 0.$$

*Proof.* Choose an index  $i$  with  $|k_i| = N$ . If  $k$  has a nonzero component transverse to  $i$ , set  $p = \text{sgn}(k_i)e_i$  and  $q = k - p$ . Then  $\|p\|_\infty = 1 \leq N - 1$ ,  $q \in K_N$ , and the transverse component makes  $p$  and  $q$  non-collinear. If instead  $k = \pm Ne_i$ , choose any  $j \neq i$ , set  $p = e_j$  and  $q = k - p$ . Again  $p \in K_{N-1}$ ,  $q \in K_N$ , and  $p \times q \neq 0$ .  $\square$

For non-collinear  $p, q$ , suitable divergence-free polarizations give a nontrivial quadratic Navier–Stokes interaction. Thus every new boundary mode is kinematically attached to the previous cutoff. The repository checker exhaustively verifies the constructive choice through  $N = 8$ .

**Remark 3** (What the lemma does not prove). Lemma 2 does not prove nonzero transfer at every state, because modal amplitudes may vanish and different triads may cancel. It also does not prove independence of the observation rows. Therefore it narrows but does not solve the all- $N$  earliest-order saturation conjecture. The remaining obstruction is algebraic: enough symmetry-reduced observation minors must be shown generically nonzero.

## 6 A quantitative inverse in the analytic triad subcase

The programme’s existing three-mode analytic subcase uses reduced invariants  $(x, y, z, r)$  and the coefficient

$$c_a = \frac{3}{10}, \quad |a|^2 = 1.$$

Its first shell-energy derivative obeys

$$J_{1,a} = 2c_a r - 2\nu x. \tag{6}$$

Therefore

$$r = \frac{J_{1,a} + 2\nu x}{2c_a}. \tag{7}$$

If  $x$  and  $J_{1,a}$  have certified absolute radii  $\sigma_x$  and  $\sigma_J$ , then

$$|r - \widehat{r}| \leq \frac{\sigma_J + 2\nu\sigma_x}{2|c_a|}. \tag{8}$$

At  $\nu = 1/200$  this simplifies to

$$|r - \widehat{r}| \leq \frac{5}{3}\sigma_J + \frac{1}{60}\sigma_x.$$

The exact rational checker uses a perturbed sample and verifies an error of 49/60000 against the certified bound 17/20000.

This is a genuine quantitative inner-inverse witness, but only for the reduced triad model. It is not a certified inversion of the 52-real-dimensional  $N = 1$  cube.

## 7 Relation to the observable-to-continuum certificate

The companion EPSC synthesis gives the conditional composition

$$\inf_{g \in G} \|u(T) - g\hat{x}_N\|_2 \leq \sqrt{\rho_N^2 + \beta_N^2},$$

where  $\rho_N$  is a certified retained-state reconstruction radius and  $\beta_N$  is an independently certified omitted-tail radius. The present note advances the inner side in two ways: it supplies a quantitative  $\rho$ -type bound in the exact reduced triad case, and it supplies derivative-free window transfer data with explicit uncertainty radii. Neither result manufactures the full  $\rho_N$  needed for a general finite Fourier state.

## 8 Updated open problems

The work leaves four deliberately separated frontiers.

1. **Full finite inverse (EPSC-18).** Fix the spatial-translation gauge, control observation branches, and certify a quantitative inverse radius for the complete finite Fourier quotient. The triad result (8) is only a low-dimensional witness.
2. **Full noisy chain (EPSC-19).** Propagate certified measurement-window errors through the full finite inverse and then through the EPSC tail composition. Equation (5) removes one differentiation bottleneck but not the inversion bottleneck.
3. **All- $N$  saturation (NSOBS-07).** Kinematic connectivity is supplied by Lemma 2; the remaining issue is generic nonvanishing/algebraic independence of enough symmetry-reduced observation minors.
4. **Outer-certificate scaling (EPSC-16).** Improve the independent finite-to-continuum certificate's cost and tightness at high cutoff and long horizon. This is not the same problem as observability inversion.

## 9 Conclusion

Energy transfer is useful in finite Navier–Stokes observability, but not because it secretly adds rank. Its mathematically honest role is more specific: it is the coordinate in which internal redistribution becomes explicit, the conservation dependence behind the shell-reader rank count is visible, and finite-window observations can be certified without high-order differentiation. The new all-cutoff connectivity lemma removes a structural obstruction to the all-resolution conjecture, while the reduced triad inverse gives a first explicit quantitative inner certificate. The full finite-state inverse, its noisy extension, and all- $N$  minor independence remain open and are stated as such.

**Reproducibility.** Executable checks and claim boundaries are maintained in `reproduction/checks/check_volume`, `reproduction/NS_ENERGY_TRANSFER_OBSERVABILITY_BRIDGE.md`, and the auto-generated reproduction ledger. Reusable helper algebra is maintained in the sister `information-discrete-math` repository. Toledo proposal records are provenance/status entries and are not automatically canonical theorem codes.